RAJALAKSHMI ENGINEERING COLLEGE

Muralikarthik V 250701458-2025-CSE

M2

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|---|---|
| **Started on** | Friday, 14 August 2026, 4:22 PM |
| **State** | Finished |
| **Completed on** | Friday, 21 August 2026, 3:51 PM |
| **Time taken** | 6 days 23 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int func(int arr[],int low,int high,int n)
{
    int mid=(low+high)/2;
    if(low>high)
    {
        return 0;
    }
    else
    {
        if(arr[mid]==0)
        {
        if(mid==0||arr[mid-1]==1)
        {
            return n-mid;
        }
        return func(arr,low,mid-1,n);
        }
        else
        {
            return func(arr,mid+1,high,n);
        }
    }
}
int main()
{
    int m;
    scanf("%d",&m);
    int arr[m];
    int low=0;
    int high=m-1;
    for(int i=0;i<m;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("%d",func(arr,low,high,m));
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course